

$$5) \ y = \frac{3x^2 - 3}{x^3}$$

$$6) \ y = (2x - 8)^{\frac{2}{3}}$$

$$7) \ y = -\frac{1}{5}(x - 4)^{\frac{5}{3}} - 2(x - 4)^{\frac{2}{3}} - 1$$

Critical thinking question:

8) If functions f and g are increasing on an interval, show that $f + g$ is increasing on the same interval.

9) Give an example where functions f and g are increasing on the interval $(-\infty, \infty)$, but where $f - g$ is decreasing.